YOUR TITLE

An Information Technology Capstone Project

Presented to the Faculty of Information and Communication Technology

South East Asian Institute of Technology, Inc.

National Highway, Brgy, Crossing Ruber, Tupi, South Cotabato

In Partial Fulfillment of the Requirements for the Degree of

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

(LAST NAME, FIRST NAME M.I)

MEMBERS

MEMBERS

MEMBERS

Adviser

KENMARK T. CELIS, LPT

September 2026

Table of Contents

CHAPTER I

INTRODUCTION

Background of the study

The Background of the Study (found in Chapter I: Introduction) follows a four-paragraph flow that moves from the general topic, to the specific setting, to the problem, and finally to your proposed solution. This structure makes the logic of 'why this study exists' clear and easy to follow.

Overall Structure (on this part the background of the study should have 2-3 page)

1. Paragraph 1 — Brief description of the study.
2. Paragraph 2 — The company/establishment and the process(es) they currently do.
3. Paragraph 3 — The problem(s) they encountered in that process.
4. Paragraph 4 — The solution, based on the features of your proposed system.

Paragraph 1: Brief Description of the Study

Open with a short overview of the project itself — this orients the reader before you introduce the setting. Keep it to 2–4 sentences and include:

- The full title of the study.
- The type of system/project it is (e.g., web-based, QR-code-based, mobile app).
- The institution or beneficiary it was made for.
- The core purpose it serves — in one sentence.

> **TIP:** Keep this brief — it's a preview, not a summary of the whole chapter.

"This study presents the Payroll, Employee Management, and Daily Time Record System Using QR Code, a computerized system developed for Dr. Francisco C. Parcon Clinic and Hospital. It is designed to automate employee attendance monitoring and payroll computation through QR code scanning, replacing the institution's manual recording methods."

Paragraph 2: The Company/Establishment and Its Process

Introduce the institution where the study was conducted. Describe what it does and, specifically, how it currently carries out the process your system will affect (e.g., how attendance is recorded, how payroll is computed).

- Brief background of the establishment: what it is, its purpose, its scale (e.g., number of employees, years in operation).

- A step-by-step description of their CURRENT process — how things are done today, before your system.
- Keep this factual and neutral; save judgment/criticism for Paragraph 3.

> **TIP:** This paragraph is often built from your interview or observation data — describe the actual workflow you observed, not an assumed one.

> "Dr. Francisco C. Parcon Clinic and Hospital is a private healthcare institution that employs medical and administrative staff to serve patients in the community. To monitor employee attendance, the institution currently relies on logbooks wherein employees manually write their time-in and time-out. At the end of each pay period, the administrative staff manually reviews these logbooks and computes each employee's rendered hours and corresponding wages using a calculator and spreadsheet."

Paragraph 3: The Problem(s) Encountered

Now shift to the weaknesses of that current process. This paragraph should make the reader feel the problem is real and worth solving.

- Identify specific issues: errors, delays, inefficiency, security risks, lost records, tampering, etc.
- If possible, support the claim with data from your interviews/questionnaires (e.g., 'X out of Y respondents reported...').
- End by stating the consequence of the problem (e.g., delayed payroll release, employee complaints, administrative burden).
- Use (1),(2) to easily to identify the problem

> **TIP:** Every problem you state here should be one your system will actually solve in Paragraph 4 — don't introduce an issue your system doesn't address.

> "(1) This manual process, however, has resulted in recurring issues for the institution. (2) Handwritten logs are prone to errors, illegible entries, and even 'buddy punching,' where an employee logs a time entry on behalf of an absent co-worker. (3) The manual computation of rendered hours and wages is also time-consuming and susceptible to human error, often leading to delays in payroll release and discrepancies in employee pay. During interviews, the administrative staff confirmed that these inaccuracies have caused confusion and dissatisfaction among employees."

Paragraph 4: The Solution (Based on System Features)

Close the Background by presenting your system as the direct answer to Paragraph 3's problems. Map each major problem to a specific feature of your system.

- Name the system and briefly restate its purpose.
- List 2–4 key features and connect each one to the problem it resolves.
- End with a short forward-looking sentence on the expected benefit/outcome.

TIP: Use a clear 'problem → feature' pairing so the reader immediately sees the cause-and-effect, e.g., 'To address [problem], the system provides [feature].'

"To address these concerns, the proposed Payroll, Employee Management, and Daily Time Record System Using QR Code will automate the attendance and payroll process. Through QR code scanning, the system eliminates buddy punching by requiring a unique, scannable employee ID for each time entry. Rendered hours and wages are computed automatically by the system, reducing human error and processing time. Administrators will also be able to monitor employee attendance and generate payroll reports in real time. With these features, the system is expected to improve the accuracy, security, and efficiency of the institution's employee management process."

Objectives of the Study

General Objectives

Specific objectives

- Develop a QR code-based time-in and time-out feature to automate attendance logging

GUIDE: WRITING THE OBJECTIVES OF THE STUDY

General Objective and Specific Objectives

The Objectives of the Study state exactly what your project intends to accomplish. They are split into one General Objective (the overall purpose) and several Specific Objectives (the concrete steps that achieve it). Together, they act as a checklist that Chapter IV's Conclusion must answer point by point.

How the Two Relate

Think of the General Objective as the destination and the Specific Objectives as the road that gets you there. Every specific objective should contribute directly to achieving the general objective — nothing more, nothing less.

A. General Objective

A single, broad sentence that summarizes the overall purpose of the study. It is usually written first and answers the question: 'What is this study meant to achieve, in one sentence?'

- Start with a fixed phrase such as 'This study aims to...' or 'This project seeks to...'
- Name the system/project and its main function.
- Name the institution/beneficiary it serves.
- Keep it to one sentence — save the details for the specific objectives.

TIP: If your General Objective sentence is more than 3 lines long, it is probably trying to do the job of a Specific Objective — trim it down.

"This study aims to design and develop a Payroll, Employee Management, and Daily Time Record System Using QR Code for Dr. Francisco C. Parcon Clinic and Hospital to automate employee attendance monitoring and payroll processing."

B. Specific Objectives

A numbered list that breaks the General Objective down into concrete, achievable, and measurable actions. Each one should be something you can later point to and say 'yes, this was accomplished' in your Conclusion.

- Start each item with a strong action verb (see verb table below) — never with vague verbs like 'know' or 'understand.'
- Each objective should describe ONE clear action or deliverable — don't combine two goals with 'and' into a single item.
- Arrange them in logical order — usually the order the system/process is built or used (e.g., design → develop → test → evaluate).
- Keep the number of specific objectives manageable — typically 3 to 6 items.

- Make sure each one is realistically achievable within your project's scope and timeline.

TIP: A useful test: could a panel member ask 'did you achieve this?' and get a clear yes/no answer just by reading your objective? If not, it's too vague.

"Specifically, this study aims to: 1. Design a database to securely store employee information, attendance records, and payroll data. 2. Develop a QR code-based time-in and time-out feature to automate attendance logging. 3. Automate the computation of employee rendered hours, deductions, and wages. 4. Provide administrators with a reporting feature to monitor employee attendance and generate payroll reports. 5. Test and evaluate the system's functionality, accuracy, and usability based on ISO 25010 quality standards."

Verb Reference Table

Use this table to choose stronger, more measurable verbs for each objective.

Category	Weak / Vague Verbs (avoid)	Strong / Measurable Verbs (use)
Creating	know, understand, learn	design, develop, create, build
Improving	help, improve (alone)	automate, streamline, enhance, optimize
Testing	check, look at	evaluate, assess, test, validate
Analyzing	study, think about	analyze, determine, identify
Delivering	make available	implement, deploy, provide, generate

Quick Checklist

- General Objective is one sentence covering the whole project.
- Every Specific Objective starts with a strong, measurable action verb.
- Each Specific Objective covers only one action/deliverable.
- Specific Objectives are arranged in logical order.
- All Specific Objectives together fully support the General Objective.
- Every Specific Objective will be revisited and answered in the Chapter IV Conclusion.

Scope and Limitation of the Study

Scope of the Study

Limitation of the Study

The Scope and Limitation of the Study defines the boundaries of your project: what the system covers (Scope) and what it does not cover (Limitation). The Scope is presented as an opening paragraph followed by a itemized list of the system's actual features — this makes it easy for panelists to see exactly what was built.

Overall Structure

1. Opening paragraph — general coverage statement (users, institution, platform type).
2. Feature list — each feature of the system named and briefly described.
3. Limitation — what the system does NOT do, stated clearly and honestly.

Part 1: Scope of the Study

Begin with a short paragraph giving the overall coverage of the system before listing individual features.

- State the platform/type of system (e.g., web-based, desktop-based).
- State who will use it (e.g., administrators, employees) and where (the institution).
- State the general functions covered, in one or two sentences, before the detailed list.

"This study covers the design and development of a web-based Payroll, Employee Management, and Daily Time Record System Using QR Code intended for use by the administrators and employees of Dr. Francisco C. Parcon Clinic and Hospital. The system covers the following features:"

Part 2: Listing Each Feature

After the opening paragraph, list every major feature of the system as a separate bullet point. Each entry should follow the format: Feature Name — a short phrase describing what it does.

- Name the feature exactly as it appears in your system (e.g., match your module names/menu labels).
- Describe its function in a single clause — avoid long explanations here (save those for Chapter III).

- List features in the order a user would encounter or use them (e.g., login → attendance → payroll → reports).
- Only include features that are actually built and functional in your system — do not list planned-but-unfinished features.

> **TIP:** This list should match exactly what's shown in your Chapter III 'List of Modules' table — if a feature is listed here, it must appear there too, and vice versa.

Sample feature list for the PEMDaS system: • QR Code Attendance Logging — allows employees to scan their ID for time-in and time-out recording. • Automated Payroll Computation — automatically calculates rendered hours, deductions, and net wages. • Employee Records Management — stores and manages employee personal and employment information. • Daily Time Record (DTR) Generation — automatically compiles employee attendance into a printable DTR. • Admin Dashboard and Monitoring — allows administrators to view real-time attendance and employee status. • Payroll Report Generation — generates printable payroll summaries and reports for a selected pay period. • User Account Management — allows the creation and management of admin and employee accounts with role-based access.

Part 3: Limitation of the Study

After presenting the Scope, write a separate short paragraph (or bullet list) clearly stating what the system does NOT do. This manages the reader's expectations and protects you during your defense.

- Technical boundaries — e.g., platforms not supported, features not automated.
- Data boundaries — e.g., government-mandated computations not included (SSS, PhilHealth, Pag-IBIG, tax tables), unless actually implemented.
- Operational boundaries — e.g., requires internet connection, limited to one branch/location, requires a QR-code-capable device.
- Time/resource boundaries — e.g., tested only within the given project timeline and sample size.

> **TIP:** Never list a limitation that contradicts a feature you listed in the Scope — panelists specifically check for this mismatch.

"This study is limited to the employee attendance monitoring and payroll computation processes of Dr. Francisco C. Parcon Clinic and Hospital only, and does not cover other human resource functions such as recruitment, leave management, or performance evaluation. The system's QR code scanning feature requires a working camera-enabled device and a stable internet connection. Government-mandated deductions (SSS, PhilHealth, Pag-IBIG, and withholding tax) are computed using fixed rates as of the time

of development and are not automatically updated to reflect future rate changes."

Quick Checklist

- Scope opens with a general coverage paragraph before the feature list.
- Every feature listed is actually built and working in the final system.
- Feature list matches the modules described in Chapter III.
- Limitation paragraph does not contradict anything listed in the Scope.
- Limitations are specific (not just 'the system has some limitations').

IPO Model

IPO (INPUT-PROCESS-OUTPUT) MODEL

Conceptual Framework / Flow of the Study

The IPO Model is the standard way to visually present 'The Flow of the Study' in Chapter I. It shows, at a glance, what data goes IN to the system, what the system DOES with it, and what comes OUT as a result.

INPUT	→	PROCESS	→	OUTPUT
• Employee personal and employment information • Employee QR code / ID • Time-in and time-out scan data • Payroll rates and deduction policies • Admin adjustments/corrections	→	• QR code scanning and validation • Attendance logging and storage • Computation of rendered hours • Payroll computation (deductions, net wages) • Data encryption and record management • Report generation	→	• Daily Time Record (DTR) • Payroll report / payslip • Employee attendance summary • Real-time admin dashboard view

HOW TO BUILD YOUR OWN IPO DIAGRAM

General Structure

An IPO diagram has three connected boxes read left to right (or top to bottom): Input → Process → Output. Each box is filled with short bullet points, not full sentences.

Box 1: Input

List everything the system needs to receive before it can do anything — raw data, user actions, or existing records.

- Data entered by users (e.g., employee records, login credentials).
- Data scanned or captured automatically (e.g., QR code scans, sensor data).
- Existing reference data the system depends on (e.g., pay rates, schedules).

> ***TIP:*** *If an item on your list doesn't come FROM outside the system, it doesn't belong in Input — it belongs in Process.*

Box 2: Process

List what the system DOES to the inputs — the actions, computations, or transformations performed internally.

- Use verbs: validate, compute, encrypt, sort, generate, store, match.
- Match each process item to a feature from your Scope of the Study — every listed feature should have a corresponding process step.
- Keep each item short (one phrase); this is a summary diagram, not the full methodology.

Box 3: Output

List what the system produces or displays as a result — the tangible deliverables the user receives.

- Reports, records, or documents generated (e.g., DTR, payslip).
- Views or dashboards presented to the user.
- Any decision or status the system determines (e.g., 'approved/denied,' 'present/absent').

> ***TIP:*** *Every Output item should trace back to an Input + Process pairing — if you can't explain how an output was produced, the diagram has a gap.*

Optional: Feedback Loop

Some studies add a dashed arrow from Output back to Input, representing continuous improvement (e.g., admin corrections feeding back into the system, or evaluation results informing future updates). Only include this if your study actually describes such a cycle.

Quick Checklist

- Each box uses short bullet points, not paragraphs.
- Input items are things that come INTO the system from outside.
- Process items are actions the system performs, matching your listed features/modules.
- Output items are what the user actually receives or sees.
- The diagram is followed by 1-2 short paragraphs in the text explaining the flow, for readers who want detail.

Definition of Terms

The **Definition of Terms** section is a standard part of a capstone project (usually placed at the end of Chapter 1 or in a preliminary section) that clarifies how key words, technical concepts, and jargon are used *specifically within your study*. It ensures readers interpret your terminology the same way you intend, especially when a word could have multiple meanings or is unfamiliar to a general audience.

Purpose

- **Clarity** – prevents misinterpretation of specialized, technical, or field-specific terms.
- **Consistency** – ensures the term is used the same way throughout the paper.
- **Accessibility** – helps readers outside your field (e.g., panel members, future researchers) understand your work.
- **Precision** – shows you understand the theoretical or conceptual grounding of your study.

What to Include

1. **Technical/field-specific terms** — words unique to your discipline (e.g., "machine learning," "phenomenology," "net present value").
2. **Variables from your study** — especially independent/dependent variables, since these need operational definitions.
3. **Acronyms and abbreviations** — spelled out and explained.
4. **Terms used in a specialized or non-standard way** — words that have a different everyday meaning versus how you use them in the study.
5. **Locally or contextually specific terms** — e.g., institution-specific programs, local terminology.

Two Common Types of Definitions

- **Conceptual (Theoretical) Definition** – the term's meaning based on literature, dictionaries, or established theory.
- **Operational Definition** – how the term is specifically measured, applied, or used *within your study* (most capstones require this, especially for variables).

Formatting Guidelines

- List terms **alphabetically**.
- Bold or italicize the term, followed by its definition.

- Keep definitions **concise** (1–3 sentences).
- Cite sources if the definition is borrowed from literature (APA/MLA depending on your institution's style).
- If a term has an operational definition specific to your study, label it clearly (e.g., *"In this study, ___ refers to..."*).

Example Format

Academic Performance – In this study, academic performance refers to the students' cumulative grade point average (GPA) at the end of the school year, as reflected in official school records.

E-Learning – A form of education delivered through digital platforms and the internet, allowing instruction outside a traditional classroom setting (Smith, 2020).

Process for Creating Chapter II:

1. List 5-8 sub-topics that your system touches.
2. Find 3-5 recent studies (local and foreign) per sub-topic.
3. Write one sub-heading per sub-topic, grouping agreeing studies into a single synthesized sentence.
4. End each sub-heading with a "However..." sentence stating the gap.
5. Combine all the gaps into one Synthesis paragraph.
6. Close with "Therefore, this study addresses these limitations by developing [your system]..." to bridge into Chapter III.
7. Build the Reference list only from sources actually cited, in APA format.
8. Also add if there is an AI in the system and the SDGs align with the system

EXAMPLE

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the related literature and studies relevant to the system. It includes both local and foreign sources that support the development of the study and provide background information.

Digital Record-Keeping in Local Government Units

The digitalization of records in local government units (LGUs) has become an increasingly important response to the inefficiencies of manual, paper-based administration. Barangay offices, as the smallest units of local governance, traditionally rely on handwritten logbooks and physical filing systems for maintaining resident records, blotter reports, and certifications, a practice widely recognized as slow, error-prone, and vulnerable to loss or damage (Baycongan & Cruz, 2022; Villaruel et al., 2021). In response, various computerized and web-based record-management systems have been developed to automate data storage, retrieval, and reporting processes within barangay and municipal offices (Domingo et al., 2023; Fernandez & Aquino, 2022).

Empirical studies show that digitized record-keeping significantly improves administrative efficiency. Systems that centralize resident and household data allow barangay personnel to retrieve records within seconds rather than searching through physical files, reducing processing time for common transactions such as certificate requests and clearance issuance (Domingo et al., 2023; Bayongan & Cruz, 2022). Similarly, cloud-based recordkeeping platforms enable data backup and remote access, minimizing the risk of permanent data loss during calamities or office relocation (Fernandez & Aquino, 2022; Villaruel et al., 2021).

However, despite these advancements, many existing barangay record-keeping systems remain narrow in scope. Several are designed only for a single function, such as resident registration or blotter recording, without integration into a unified information system (Domingo et al., 2023; Fernandez & Aquino, 2022). Moreover, a number of these systems remain at the pilot or prototype stage,

implemented in only one or two barangays, limiting evidence of their scalability to larger or more diverse local government contexts (Villaruel et al., 2021; Bayongan & Cruz, 2022).

Resident Profiling and Certificate Issuance Systems

Accurate resident profiling forms the foundation of effective barangay governance, as it supports service delivery, planning, and the issuance of official documents such as barangay clearances, indigency certificates, and business permits. Digital profiling systems aim to replace manual household surveys and card-based filing with structured electronic databases that capture demographic, health, and livelihood information (Ramos et al., 2022; Adeyemi & Okoro, 2021).

Research demonstrates that automated certificate issuance modules substantially reduce transaction time and human error. Systems that generate clearances and permits directly from a resident database eliminate repetitive manual encoding, allowing barangay staff to process requests more quickly and consistently (Ramos et al., 2022; Santos & Villamor, 2023). In addition, profiling systems integrated with reporting dashboards allow barangay officials to generate demographic summaries for use in local planning and disaster preparedness (Adeyemi & Okoro, 2021; Santos & Villamor, 2023).

Nevertheless, the effectiveness of these systems is often constrained by data quality and adoption challenges. Studies note that inconsistent or outdated resident data, arising from irregular updating practices, undermines the reliability of profiling systems (Santos & Villamor, 2023; Ramos et al., 2022). Furthermore,

several systems lack safeguards against duplicate or fraudulent certificate requests, exposing gaps in verification and accountability (Adeyemi & Okoro, 2021).

Community Communication and Notification Systems

Timely communication between barangay officials and residents is essential for disseminating public advisories, emergency alerts, and community announcements. Traditional methods such as public postings and door-to-door notices are often slow and limited in reach, prompting the development of digital notification platforms for local governance (Marasigan & Del Rosario, 2022; Chukwu et al., 2021).

Studies indicate that SMS-based and mobile application notification systems considerably improve the speed and reach of barangay communication. Platforms that automatically send alerts for community events, health advisories, or emergency situations enable residents to receive information in real time, improving public safety and civic participation (Marasigan & Del Rosario, 2022; Chukwu et al., 2021). Web-based announcement boards further support transparency by allowing residents to view ongoing projects, meeting schedules, and public notices at any time (Chukwu et al., 2021; Marasigan & Del Rosario, 2022).

However, these systems face adoption barriers similar to those found in other local-government platforms. Limited internet connectivity and low smartphone penetration in some barangays restrict the reach of digital

notifications, particularly among older residents (Chukwu et al., 2021; Marasigan & Del Rosario, 2022). Additionally, many notification systems function only as one-way broadcast tools, offering little to no mechanism for residents to respond, request assistance, or report concerns back to barangay officials.

Data Security and Access Control in Government Information Systems

As barangay information systems increasingly store sensitive resident data, security and access control have become critical design considerations. Government information systems are expected to comply with data privacy regulations while remaining accessible to authorized personnel across different administrative roles (Nolasco et al., 2023; Owusu & Mensah, 2022).

Research highlights those role-based access control mechanisms, which restrict system functions according to user privileges, significantly reduce the risk of unauthorized data modification or disclosure (Nolasco et al., 2023; Owusu & Mensah, 2022). Encrypted data storage and audit-trail logging further strengthen accountability by allowing administrators to track changes made within the system (Owusu & Mensah, 2022).

Despite these safeguards, several studies report that smaller local-government systems often implement only minimal security measures due to limited technical expertise and budget constraints (Nolasco et al., 2023). This leaves resident data vulnerable to unauthorized access, particularly in systems that lack proper authentication protocols or regular security audits (Owusu & Mensah, 2022; Nolasco et al., 2023).

System Limitations and Usability Challenges in Existing Platforms

Despite the growing number of digital systems developed for barangay governance, several recurring limitations are evident across existing studies. A prominent issue is the lack of integration between core functions such as resident profiling, certificate issuance, and community communication, resulting in fragmented systems that require separate logins or platforms for each task (Domingo et al., 2023; Marasigan & Del Rosario, 2022).

Another significant limitation is limited two-way engagement between residents and barangay officials. Most existing platforms function primarily as administrative tools for barangay staff, offering residents little to no ability to track their own requests, submit concerns, or receive personalized updates (Ramos et al., 2022; Chukwu et al., 2021).

In addition, many systems provide minimal reporting and analytics capabilities, limiting officials' ability to generate demographic insights or monitor service-delivery trends over time (Adeyemi & Okoro, 2021; Santos & Villamor, 2023). Issues of scalability and security also persist, as several platforms remain confined to single-barangay pilot deployments with limited safeguards for resident data (Nolasco et al., 2023; Villaruel et al., 2021).

Synthesis of the Literature and Research Gap

The reviewed literature highlights the expanding role of digital systems in improving barangay governance, resident service delivery, and community

communication. Existing platforms have demonstrated meaningful contributions in reducing processing time, improving data accessibility, and strengthening transparency by replacing manual, paper-based processes with centralized digital solutions (Domingo et al., 2023; Bayongan & Cruz, 2022).

However, consistent limitations remain across these systems. Many platforms operate within narrow functional scopes, addressing isolated components such as resident profiling or certificate issuance without integrating them into a comprehensive system (Domingo et al., 2023; Ramos et al., 2022). The lack of two-way communication channels further limits resident engagement, while inconsistent security practices leave sensitive data vulnerable in several implementations (Chukwu et al., 2021; Nolasco et al., 2023). Additionally, several systems emphasize administrative processing over analytical insight, offering limited reporting features that could otherwise support local planning and decision-making (Adeyemi & Okoro, 2021; Santos & Villamor, 2023).

These gaps highlight the need for a more holistic and resident-centered approach to barangay information management. Specifically, there is limited development of platforms that integrate resident profiling, certificate issuance, community notification, and secure data management into a single, accessible system.

Therefore, this study addresses these limitations by developing the Barangay Information and Management System (BIMS), a web-based platform designed to provide a comprehensive, secure, and user-friendly solution for

barangay record-keeping and community service delivery. The system aims to streamline administrative processes, strengthen resident engagement, and support data-driven local governance by integrating key barangay functions into a unified platform.

CHAPTER III

METHODOLOGY

The framework of the system outlines the steps involved in developing the Coconut Processing Monitoring System. It primarily includes 4 phases: requirements, design and development, testing, and implementation. The system comprises the necessary software requirements. In the software development phase, coding plays a key role. This phase focuses on building the mechanism. After these phases, user testing is conducted to identify potential improvements and revisions for integration. The desired output is an initial system that users can start to operate, with room for future enhancements based on feedback and testing.

Environment

The setting for this study will be at the Tupi Supreme Activated Carbon, Inc., Barangay Kablon, Tupi, South Cotabato,Tupi, South Cotabato.

Software Engineering Technology

The incremental agile methodology is a way to build the Coconut Processing Monitoring System Tupi Supreme Activated Carbon, Inc. This approach allows for ongoing improvements and changes based on users' needs. Each part goes through steps like designing and developing, testing, and implementation, with user input. This process helps create a better product that meets user needs and identifies risks early, making the final system more user-friendly.

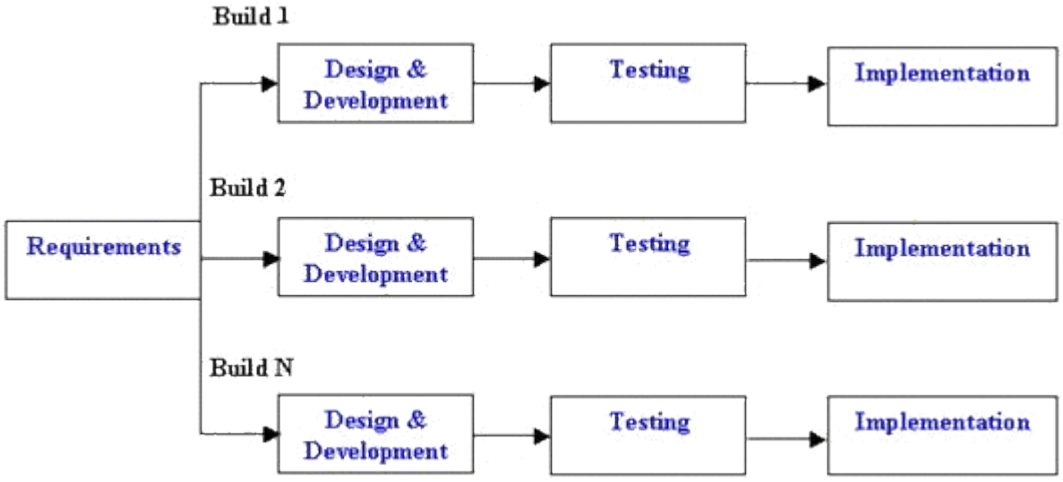


Figure 3.1: Incremental Agile Software Development Methodology Model of Coconut Processing Monitoring System

The researchers aim to find the core problems and constraints that occur in the running system and come up with a project title, as well as objectives, scope, and time constraints.

Build 1 (Provide an automated inventory management feature to ensure accurate monitoring of coconut buying and weighing)

The system aims to create an automated inventory management feature to ensure accurate monitoring of coconut buying and weighing. In the design phase, researchers will identify an accurate monitoring of coconut buying and weighing to be implemented. A detailed project plan will outline accurate monitoring of coconut buying. The architecture will be planned to utilize PHP for backend processing and HTML/CSS for the frontend. The user interfaces will be designed to be intuitive, ensuring thatcan easily to monitoring of coconut buying. Simultaneously, backend specifications will be established to handle user authentication, complemented by a well-structured database schema for storing user data and availability-related information.

In the testing phase, the system will undergo rigorous evaluation to ensure functionality and reliability. Unit tests will verify the accuracy of individual components, ensuring that continuous transactions occur between the backend and frontend. Functional testing will assess key features, like the inventory management feature, to ensure accurate monitoring of coconut buying and weighing. Throughout this phase, user feedback will be actively collected to identify issues and areas for enhancement.

The implementation of the system will occur in incremental stages, allowing for the gradual deployment of features. User acceptance testing (UAT) will involve an automated inventory management feature to ensure accurate monitoring of coconut buying, leading to further refinements. Additionally, user tutorials will be prepared to guide users through the platform, enhancing their experience and ensuring they can navigate the system effectively. This iterative approach will

enable ongoing improvements based on user feedback, ultimately leading to a strong and user-friendly platform.

Build 2 (Provide a digital recording system for raw material acquisition that provides real-time updates on stock levels)

The functionality for generating a digital recording system for raw material acquisition that provides real-time updates on stock levels. This includes identifying key input requirements, such as providing real-time updates on the stock levels module. The researchers will plan the overall architecture of the system, incorporating both backend and frontend components. The backend will be responsible for real-time inventory and data processing, while the frontend will offer an intuitive interface that provides real-time updates on stock levels. During this phase, a detailed project plan will outline the necessary real-time updates on stock levels, timelines, and resources for the phases. Technologies like PHP will be selected for backend processing, and Bootstrap will be used for responsive frontend design, alongside setting up the initial configurations for the development environment.

In the testing phase, the system will undergo a series of rigorous evaluations to ensure the accuracy and functionality provides real-time updates on stock levels. Unit tests will be conducted to verify that individual components of the backend logic function correctly, while integration tests will ensure the whole interaction between the backend and front end. Functional testing will confirm that all features, including viewing and requesting available consultation time.

Additionally, user acceptance testing (UAT) will involve feedback from a group of staff and admin to identify issues and areas for improvement. This iterative testing will allow for the early detection of problems and guide necessary refinements.

The implementation phase will see the gradual deployment of the view and provide real-time updates on stock levels, beginning with a prototype to demonstrate how the view and recording system for raw material acquisition and displayed to users. User tutorials will be prepared to help users navigate the system effectively. Any issues identified during UAT will be promptly addressed, and the system will be refined based on user feedback. This incremental approach allows for continuous improvement, ensuring that the final product meets user needs and functions as expected.

Build 3 (Tracking module for coconut by-products (husk, shell, meat, and water) to promote their efficient utilization and minimize waste)

The system will feature a monitoring feature that allows tracking module for the coconut by-products. This monitoring feature system will track coconut products. Leveraging MySQL's database capabilities, the system will efficiently store to track consultation sessions, with organized and synchronized data across devices through phpMyAdmin.

During the testing phase, the tracking module for coconut by-products will undergo rigorous tracking sessions to ensure the accuracy that allows administrators to track coconut by-products. Integration tests will be conducted to verify the whole interaction between the backend and frontend components, ensuring that data flows correctly and updates in real-time. Additionally, the system

will be subjected to functional testing to confirm that all features work as intended. User feedback will be actively collected to identify areas for improvement, ensuring that the system meets user expectations.

In the implementation phase, the system module that allows tracking module for coconut by-products, starting with a prototype that allows tracking of consultation session transactions. The user interface will be designed to ensure clarity and a tracking module, enabling easy exploration. Any issues identified during testing will be promptly addressed, and enhancements will be made based on user feedback. This iterative approach will ensure that it tracking module system module.

Build 4 (Provide a production yield reporting function that provides transparency on by-product performance and profitability)

The researchers will design a system that creates a production yield reporting function that provides transparency on by-product performance and profitability, and supports better decision-making and resource planning. This feature enables faculty and administrators to review past reports. An administrator to review past reports, and support better decision-making and resource planning.

In the testing phase, usability tests will be conducted to ensure that the process of exchanging a log system that records all reporting function activities module is straightforward and user-friendly. The system will undergo functional testing to verify that all features, such as the log system that records all reporting function activities, operate as intended. User feedback will be actively gathered

during this phase to identify any issues and areas for improvement, allowing the researchers to make necessary adjustments before the final implementation.

During the implementation phase, the clog system that records all reporting function activities module will be gradually rolled out, starting with a prototype. The user interface will be designed for clarity and ease of use, facilitating a smooth experience when navigating connectivity options. Any issues identified during testing will be promptly addressed, and enhancements will be made based on user feedback. This incremental approach will ensure that the final system is reliable and effectively meets the needs of the reporting function, providing them with affordable options to support their learning.

Build 5 (Provide expense monitoring and financial tracking component to record real-time operational costs)

The researchers will design the Coconut Processing Monitoring System to ensure a guaranteed expense monitoring and financial tracking component to record, ensuring better time management, and compare them with income for improved financial planning. Reliable software will be utilized to support system connectivity. The planning process will outline the necessary infrastructure, protocols, and technologies needed to maintain transactions continuously, including expense monitoring and financial tracking. The user-friendly interface will allow access to expense monitoring and a financial tracking dashboard in real-time, fostering transparency and trust in the service. The system will be developed using PHP for backend logic, with the front end crafted using Bootstrap and CSS frameworks for a responsive design. This phase will ensure that the design

supports dynamic adjustments to varying connection qualities, providing an uninterrupted experience for expense monitoring and financial tracking.

A full testing process will be implemented to validate the effectiveness of the expense monitoring and financial tracking component to record real-time dashboard features. Multiple rounds of testing will be conducted to identify and rectify potential issues, focusing on the overall functionality, capability, and accuracy of the system. Performance testing will ensure the system is consistently functioning, while user acceptance testing (UAT) will gather feedback from students to identify areas for refinement. This iterative testing will allow the team to address concerns and enhance the system before full deployment.

During the implementation phase, the Coconut Processing Monitoring System will be gradually rolled out, starting with a prototype that demonstrates its features. The user interface will be designed for clarity and ease of use, enabling an expense monitoring and financial tracking component to record a real-time dashboard. Any issues identified during the testing phase will be promptly addressed, and improvements will be made based on user feedback. By the end of this phase, the researchers aim to deliver a strong system solution that significantly enhances the expense monitoring and financial tracking component to record real-time.

Planning/Conception–Initiation Phase

Business Model Canvas

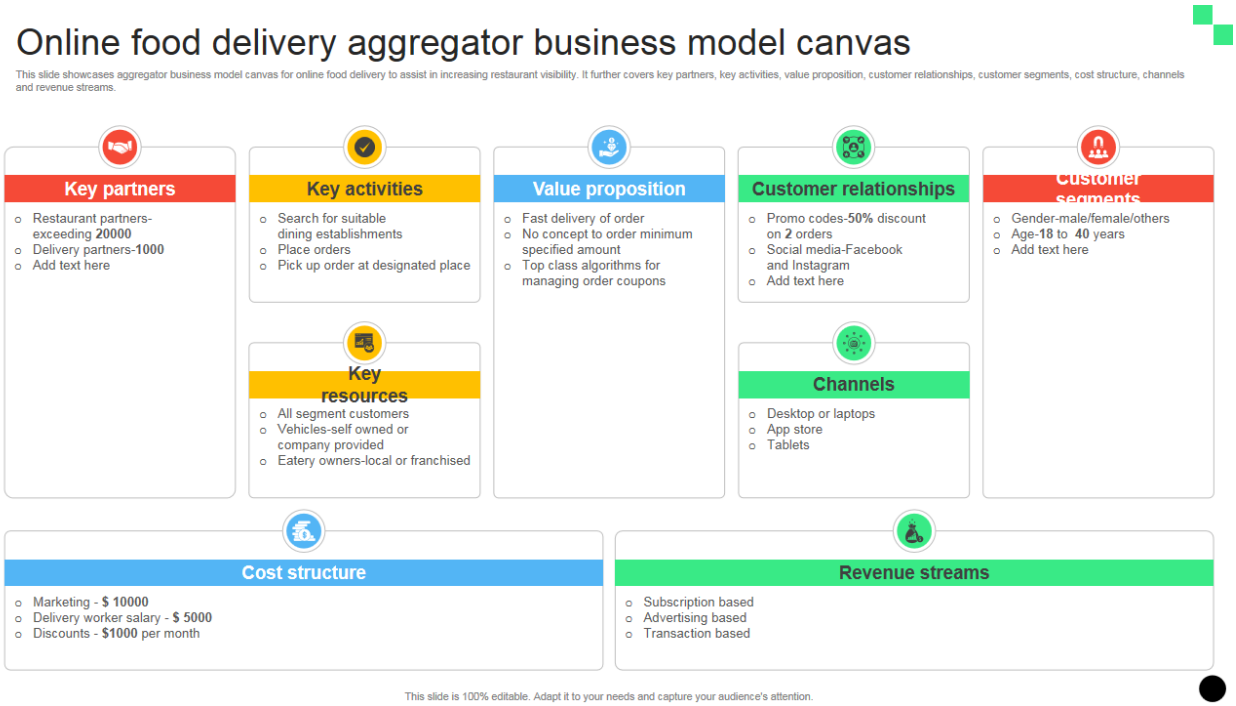


Figure 3….

Gantt Chart

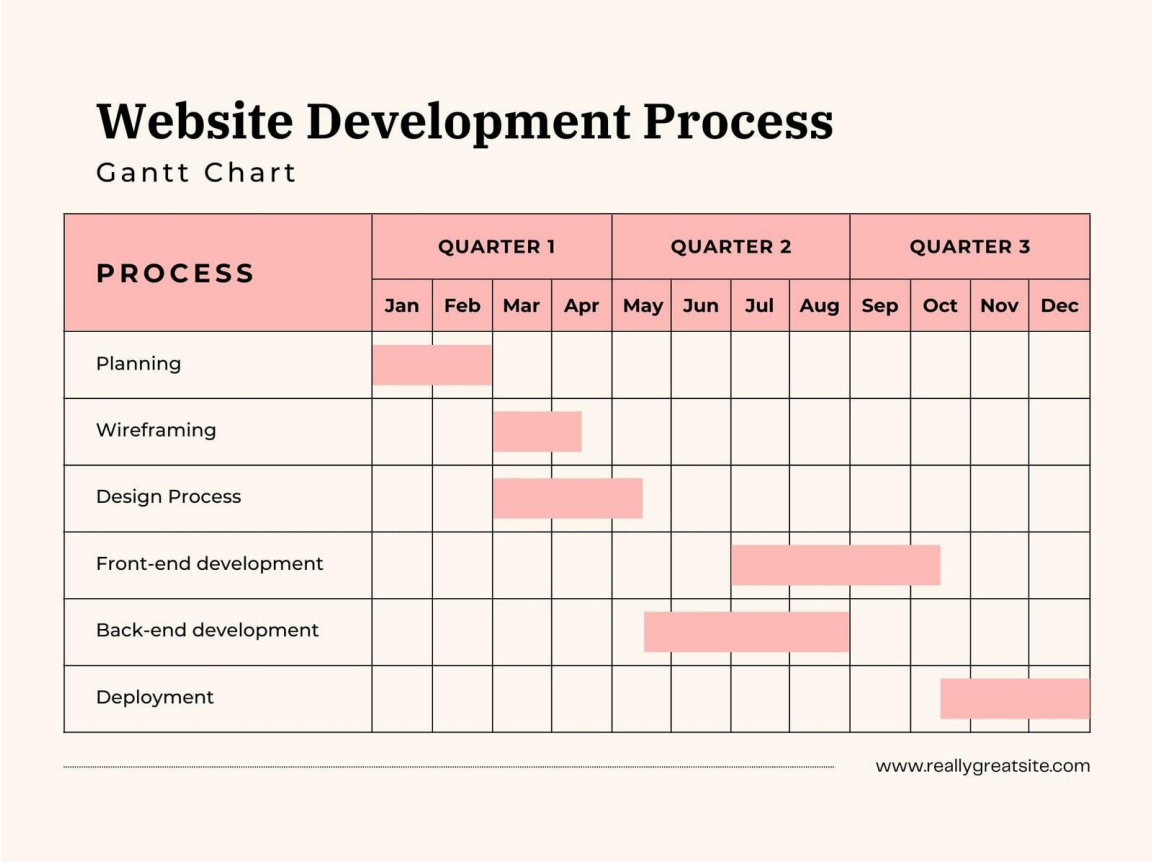


FIGURE 3…. SAMPLE LANG NI

Functional Decomposition Diagram

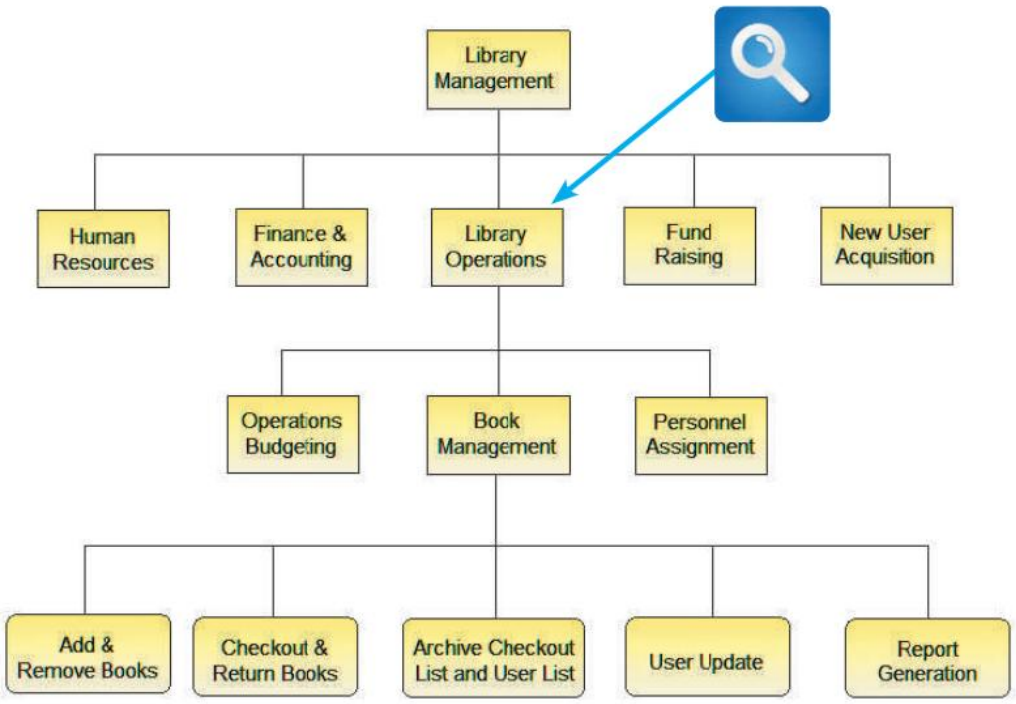


FIGURE 4-8 This Visible Analyst FDD shows a library system with five top-level functions. The Library Operations function includes two additional levels of processes and sub-processes.

Source: Visible Systems Corporation.

Figure 3…. Sample fdd

Analysis–Design Phase
Use Case Diagrams

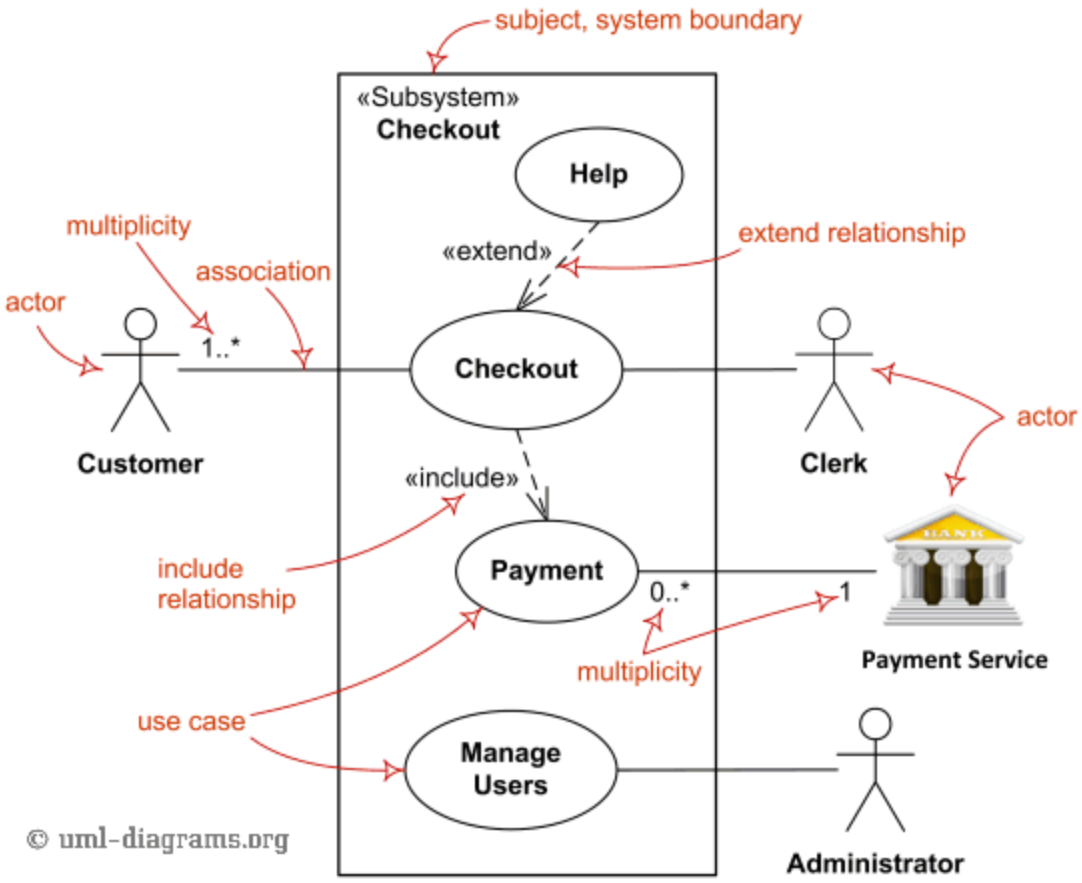


Figure 3….. Sample lang

Use Case Matrix

Use Case	Registration
Author	Bantilan, Johnrhoyd, Salarza, John Mark, Tabligan, Alex
Primary Actor	Admin
Trigger	When an admin monitor of coconut buying
Basic Flow	1. Administrators to track the tracking module for coconut by-products
	2. Monitor and financial tracking
	3. Monitor inventory
Post Condition	Admin must monitor inventory
Secondary Actor	Staff
Trigger	When staff can ensure accurate monitoring of the coconut buying and weighing
Basic Flow	1. Recording system for raw material acquisition
Post Condition	Systematic tracking module for coconut by-products (husk, shell, meat, and water)

Table 3.1: Use Case Matrix of Coconut Processing Monitoring System.

Storyboard

Admin Dashboard

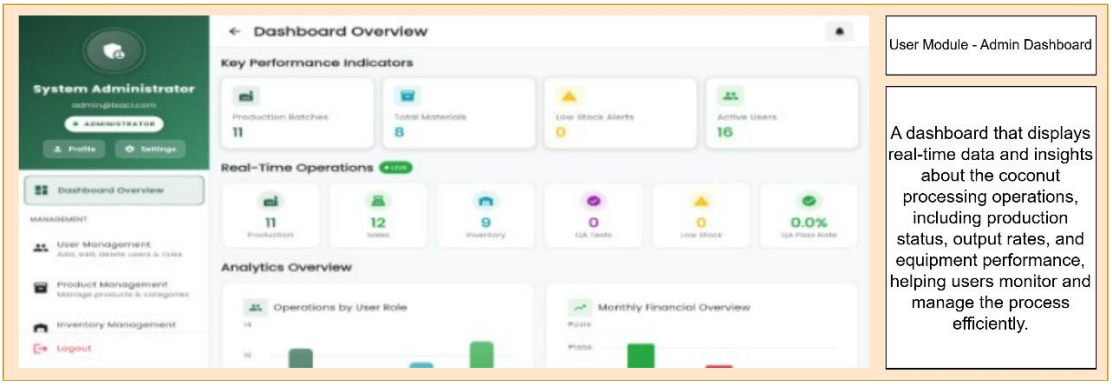


Figure 3.5.1: Use admin Dashboard of Coconut Processing Monitoring System.

Admin - User Management

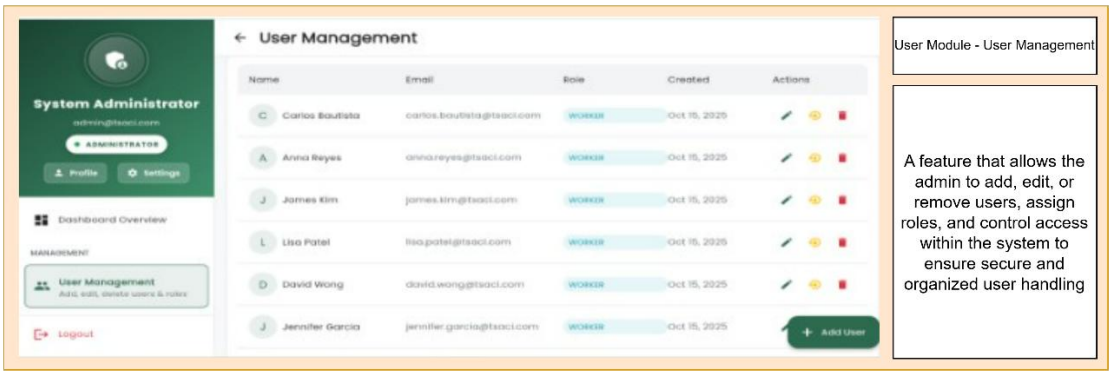


Figure 3.5.2: User Management of Coconut Processing Monitoring System.

Admin – Production Planning

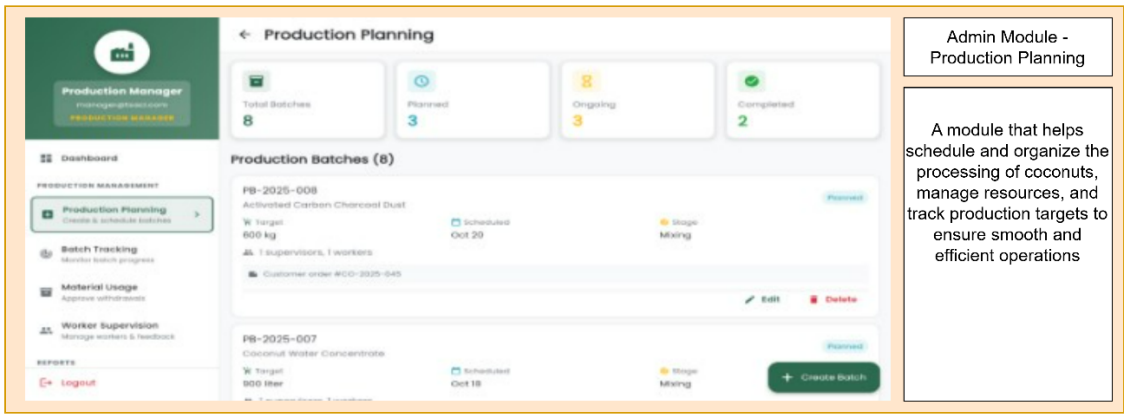


Figure 3.5.3: Production of the Coconut Processing Monitoring System.

Admin - Production Dashboard

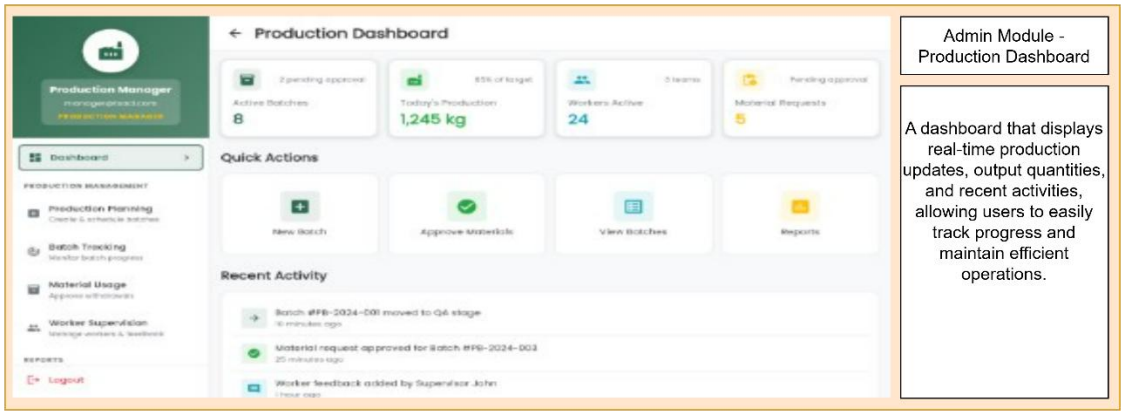


Figure 3.5.4: Production Dashboard of Coconut Processing Monitoring System.

Admin - Production Batch Tracking

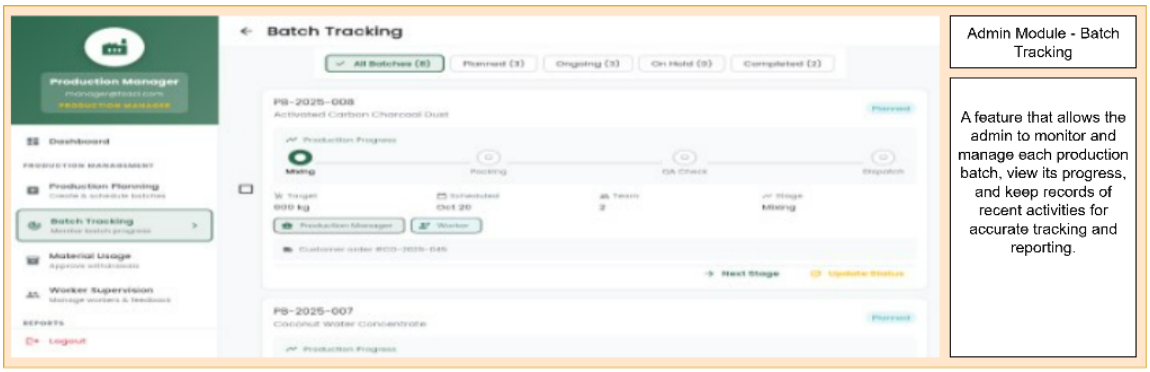


Figure 3.5.5: Production Batch Tracking of Coconut Processing Monitoring System.

Admin - Production Material Usage

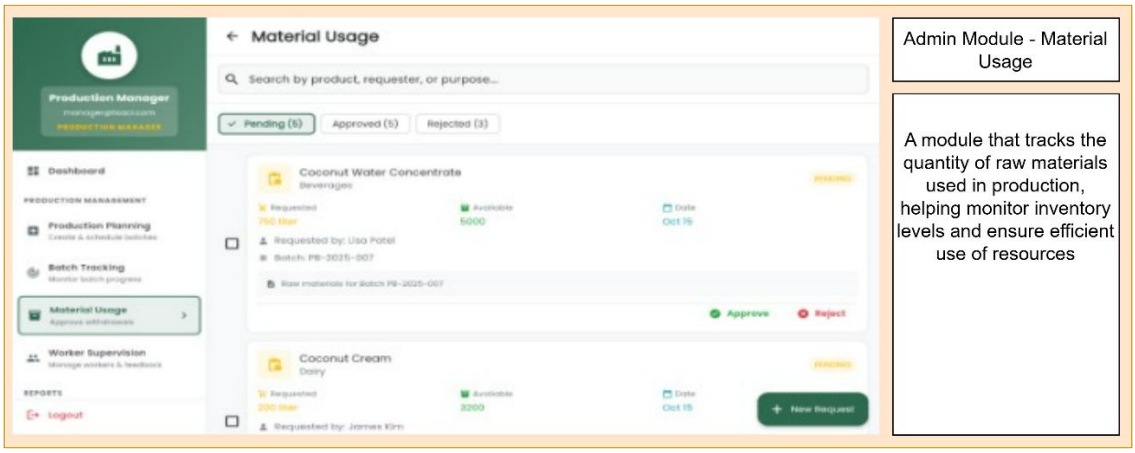


Figure 3.5.6: Production Material Usage of Coconut Processing Monitoring System.

Admin - Production Worker Supervision

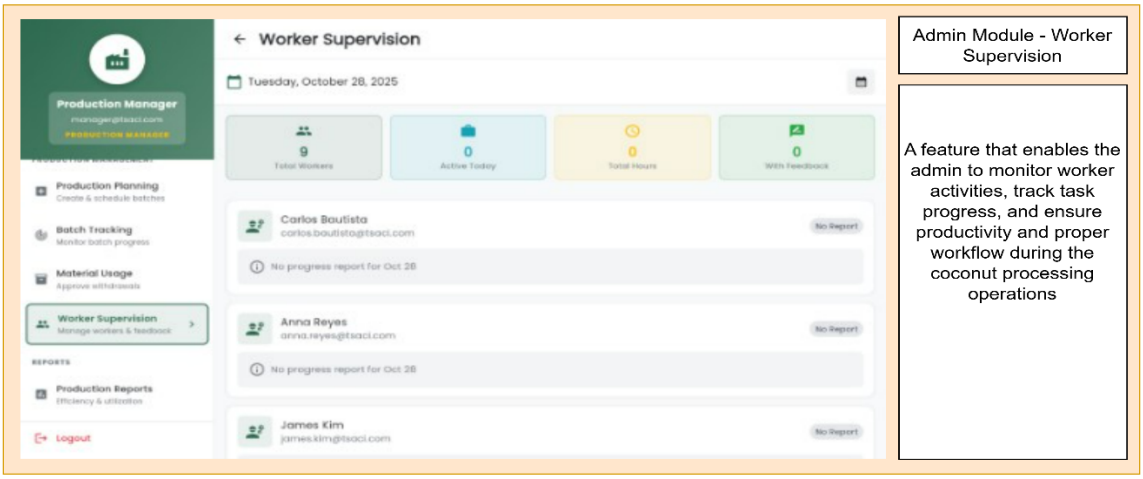


Figure 3.5.7: Production worker supervision of Coconut Processing Monitoring System.

Admin - Production Reports

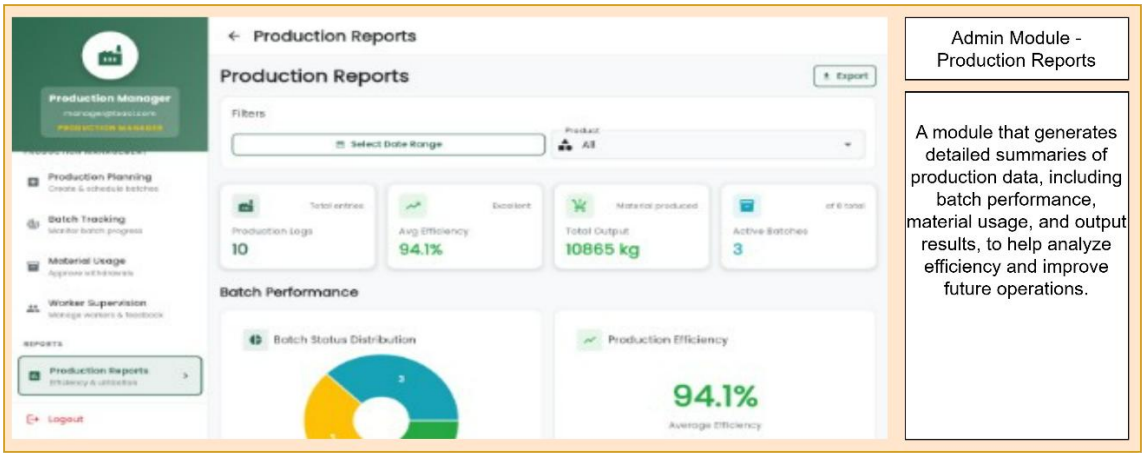


Figure 3.5..8: Production reports of Coconut Processing Monitoring System.

Inventory Dashboard

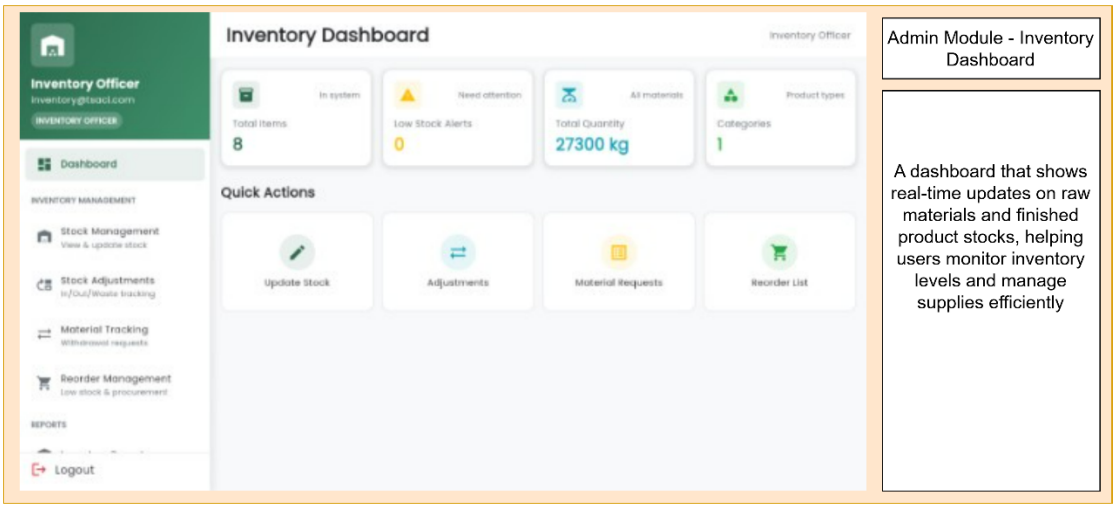


Figure 3.5..9: Inventory Dashboard of Coconut Processing Monitoring System.

Admin - Inventory Stock Management

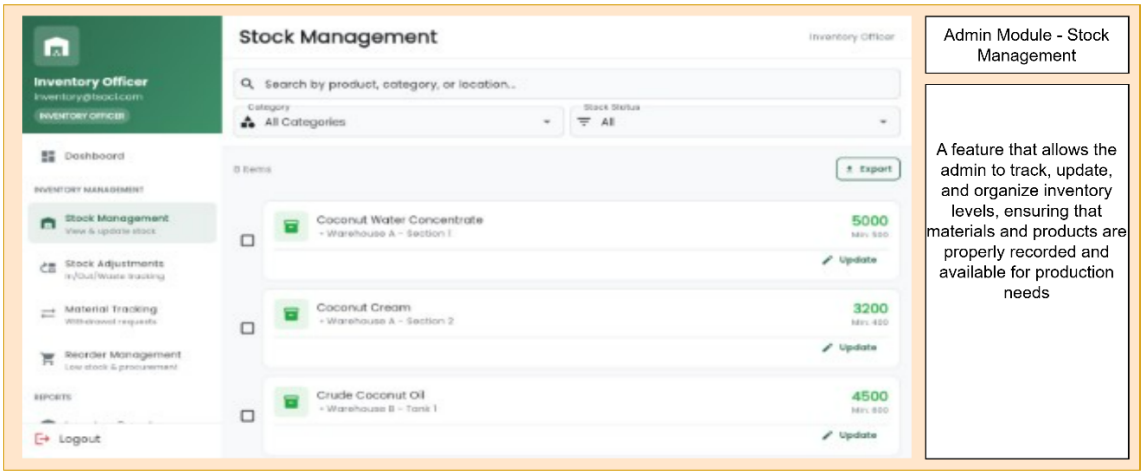


Figure 3.5..10:inventory stock managemert of Coconut Processing Monitoring System.

Admin - Inventory Stock Adjustment

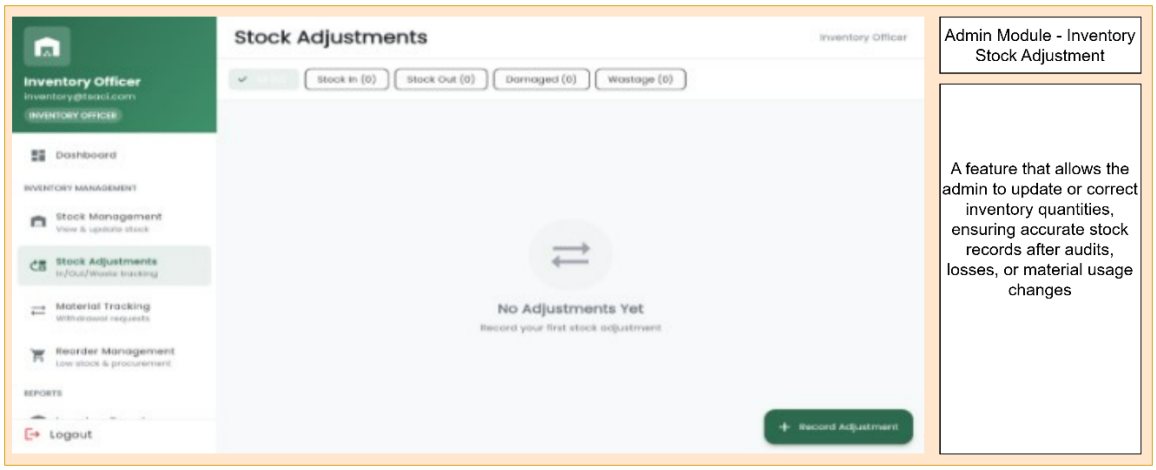


Figure 3.5..11:inventory adjustment of Coconut Processing Monitoring System.

Admin - Inventory Material Tracking

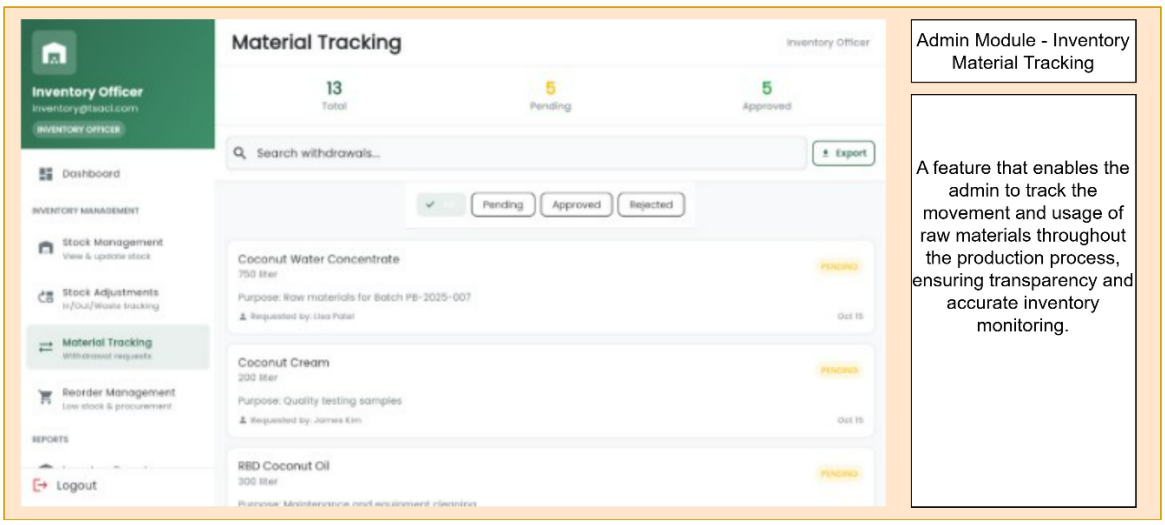


Figure 3.5.12: inventory material tracking of Coconut Processing Monitoring System.

Inventory Reorder Management

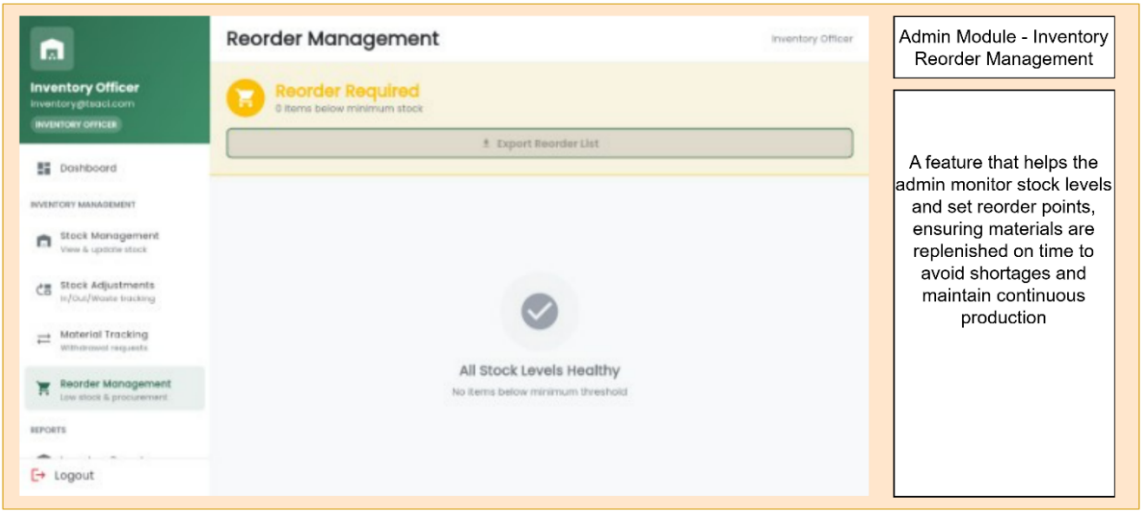


Figure 3.5.13: Inventory Reorder Management of Coconut Processing Monitoring System.

Admin - Inventory Report

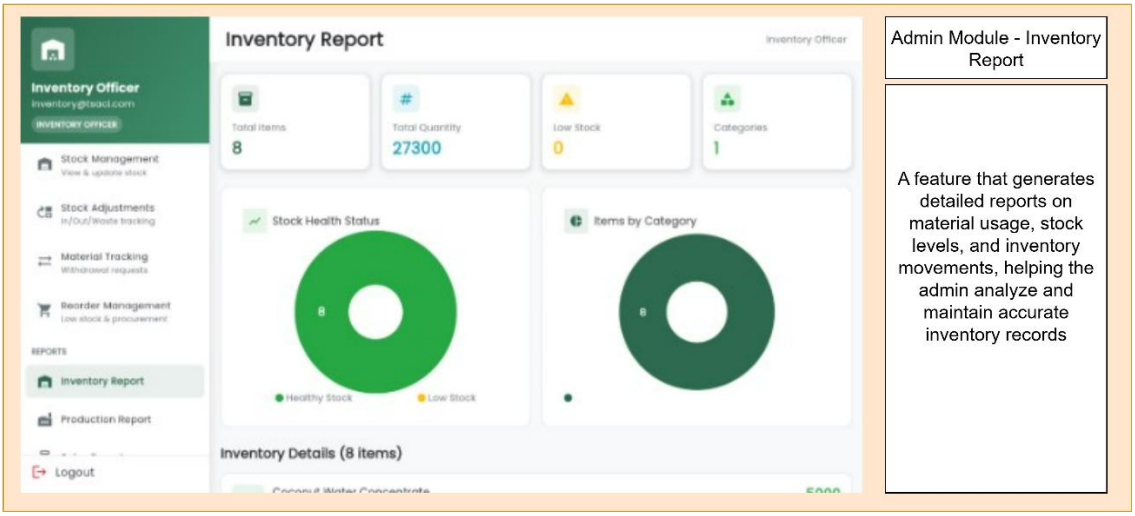


Figure 3.5.14: Inventory Report of Coconut Processing Monitoring System.

Admin - Inventory Production Report

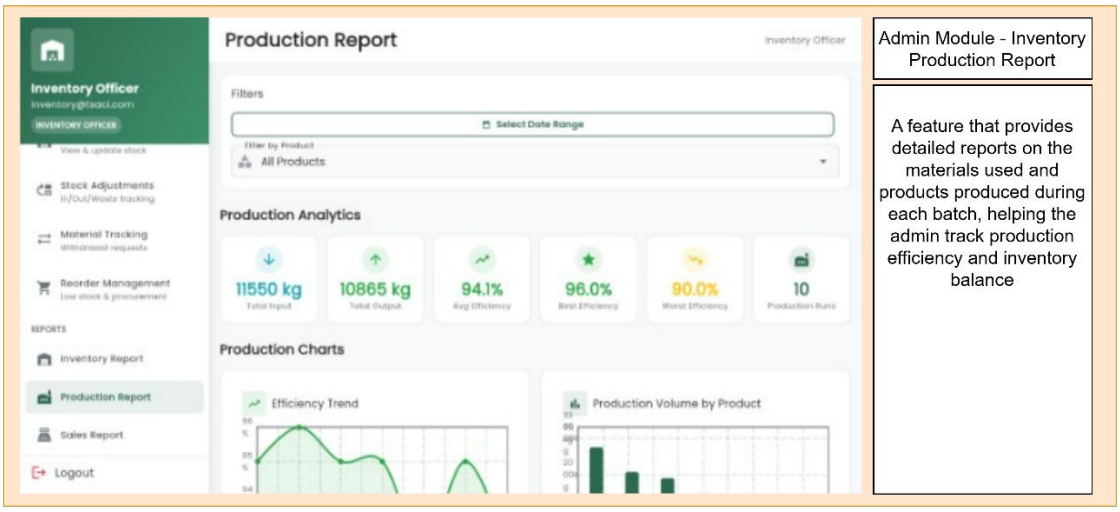


Figure 3.5.15: inventory Production report of the Coconut Processing Monitoring System.

Admin - Sales Report

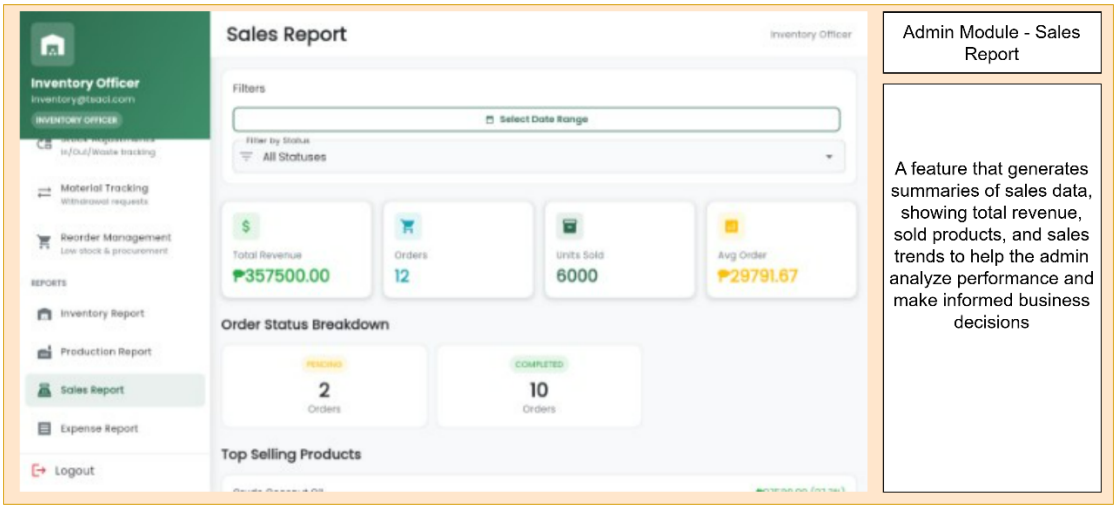


Figure 3.5.16 inventory Sales report of Coconut Processing Monitoring System.

Expense Report

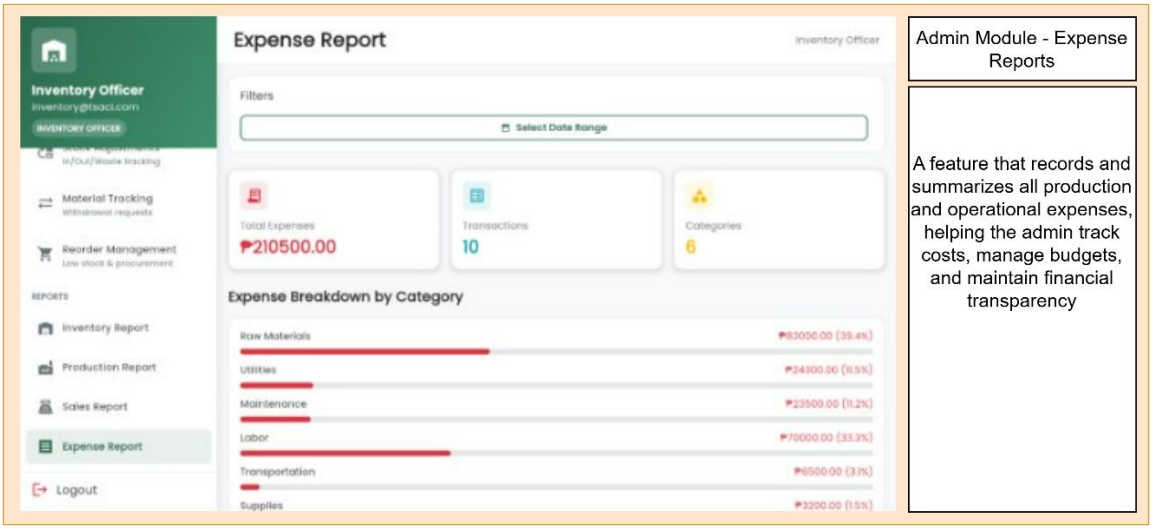
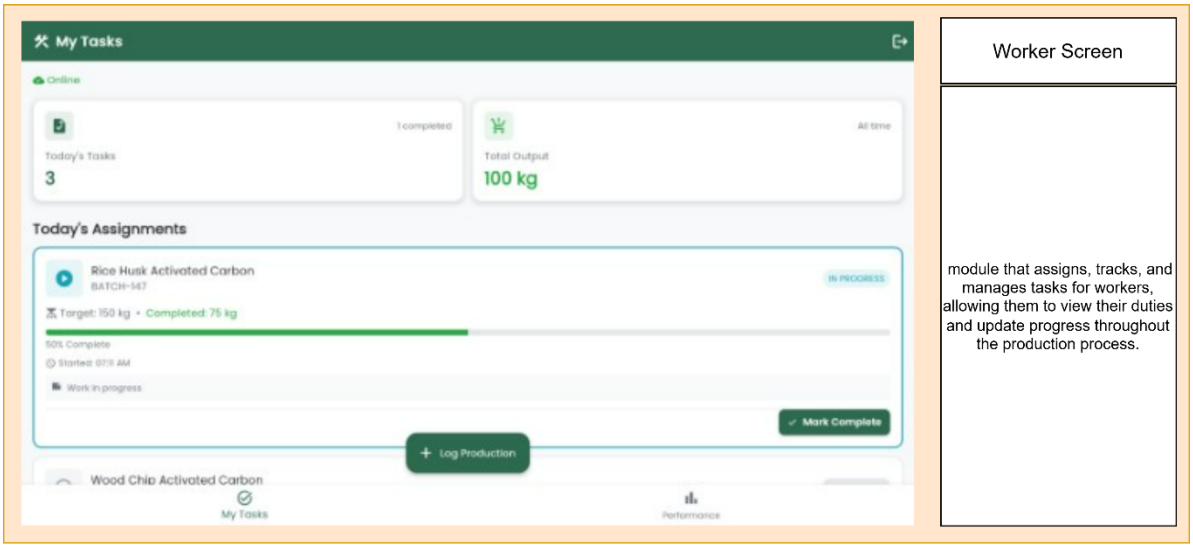


Figure 3.5.17: Expense report of Coconut Processing Monitoring System.

Worker screen

Figure 3.5.18: Worker Task screen of Coconut Processing Monitoring System.



Database Design

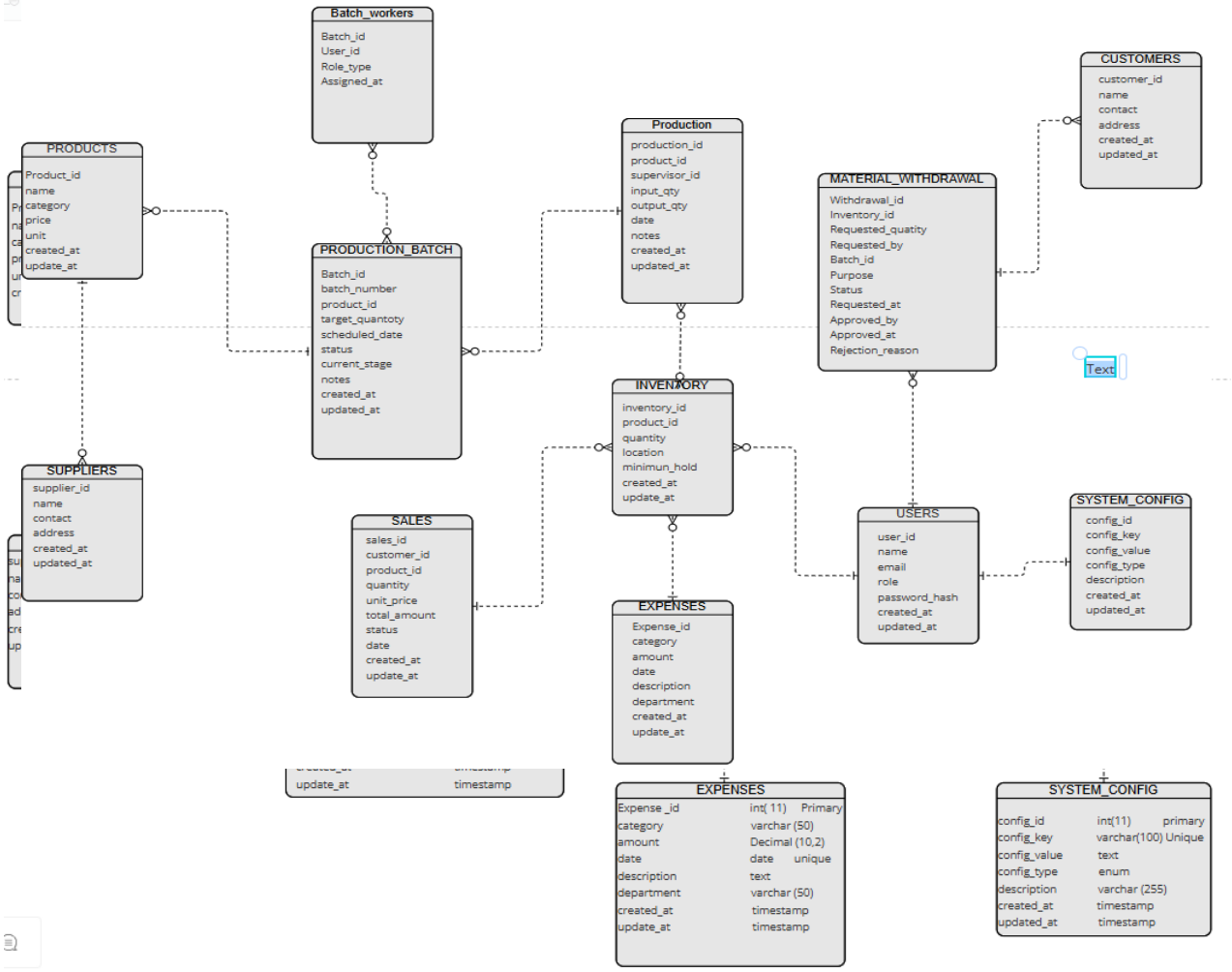


Figure 3.7: Coconut Processing Monitoring System.
Description: This figure showzs the Database Design

Entity Relationship Diagram

Figure 3.8: Coconut Processing Monitoring System.Description: This figure shows the Entity Relationship Design

Figure 3.8: Coconut Processing Monitoring System.Description: This figure shows the Entity Relationship Design

Data Dictionary

Suppliers table

Suppliers	**admin**			
Attribute	**Type**	**Null**	**Default**	**Description**
Supplier_id	INT(25)	NO	PK	Id of the supplier
Contact	VARCHAR(100)	NO		Contact of the supplier
Address	Text	NO		Address of the supplier
Created_at	Timestamp	NO		Created of supplier
Updated_by	Timestamp			Upadate of supplier

Table 3.2.1: Suppliers Table of Coconut Processing Monitoring System.

Products

Products	**Admin**			
Attribute	**Type**	**Null**	**Default**	**Description**
Product_id	INT(11)	NO	PK	id of the product
Name	VARCHAR(100)	NO		Name of product
Category	VARCHAR(50)	NO		Category of prudoct
Price	Decimal	NO		Price of the prudoct
Unit	Timestamp			Unit of the product
Created_at	Timestamp			Create of prudoct

Table 3.2.2: Products Table of Coconut Processing Monitoring System.

Batch_worker

Batch_workers	**admin**			
Attribute	**Type**	**Null**	**Default**	**Description**
Batch_id	INT(25)	NO	PK	Id of the worker
User_id	VARCHAR(100)	NO	Unique	User id of the worker
Role_type	Enum	NO		Role type of worker
Assigned_by	Timestamp	NO		Worker assigned

Table 3.2.3:Batch_worker Table of Coconut Processing Monitoring System.

Production_batch

Production_Batch	admin			
Attribute	**Type**	**Null**	**Default**	**Description**
Batch_id	INT(11)	No	PK	Id of the production
Batch_number	VARCHAR(50)	No	Unique	Number of the batch
Product_id	INT(11)	No	Unique	Id of the product
Target_quantity	Decimal (10,2)	No		Quantity of the product
Scheduled_date	Date	No		Scheduled date_of the product
Status	Enum	No	Unique	Status of the product
Current_stage	Enum	No	Unique	Stage of the product
notes	Text	No		Note of the product
Created_at	Timestamp	No		Create of the product
Updated_by	Timestamp	No		Update of the product

Table 3.2.4: Production batch Table of Coconut Processing Monitoring System.

Sales

Sales	admin			
Attribute	**Type**	**Null**	**Default**	**Description**
Sales_id	Int (11)	NO	PK	Id of the sales
Customer_id	Int (11)	NO	Unique	Customer of the sales
Product_id	Int (11)	NO	Unique	Product id of the sales
Quantity	Decimal (10,2)	NO		Quantity of the sales
Unit_Price	Decimal (10,2)	NO		Unit price of the sales
Total_amount	Decimal (10,2)	NO		Total amount of sales
Status	Emun	NO		Status of the sales
Date	date	NO		Date in of the sales
Created_at	Timestamp	NO		Create of the sales
Updated_by	Timestamp	NO		Updated of the sales

Table 3.2.5: Sales Table of Coconut Processing Monitoring System.

Production	admin			
Attribute	**Type**	**Null**	**Default**	**Description**
Production_id	INT(11)	NO	PK	Id of the production
Product_id	INT(11)	NO	Unique	Id of the product
Supervisor_id	INT(11)	NO	Unique	Id of the supervisor
Input_qty	Decimal(10,2)	NO		Input_qty of the product
Output_qty	Decimal (10,2)	NO		Output_qyt of the product

date	Date	NO	Unique	Production date
notes	Text	NO		Production notes
Created_at	Timestamp	NO		Create of the production
Updated_by	Timestamp	NO		Updated of the production

Production

Table 3.2.6: Production Table of Coconut Processing Monitoring System.

Inventory

Inventory	**admin**			
Attribute	**Type**	**Null**	**Default**	**Description**
Inventory_id	INT(11)	NO	PK	Id of the inventory
Product_id	INT(11)	NO	Unique	Id of the product
Quantity	Decima (10,2)	NO		Quantity of the product
Location	Varchar(100)		Unique	Location of the product
Minimum_Hold	Decimal(10,2)			Minimum of the inventory
Created_at	Timestamp	NO		Create of the inventory
Updated_by	Timestamp	NO		Update of the inventory

Table 3.2.7: Inventory Table of Coconut Processing Monitoring System.

Expenses

Expenses	**admin**			
Attribute	**Type**	**Null**	**Default**	**Description**
Expenses	INT(25)	NO	PK	Id of the expenses
Category	Varchar(50)			Category of the expenses
Amount	Decimal(10,2)			Amount of the expenses
Date	Date		Unique	Date of the expenses
description	Text			Description of the expenses
Department	Varchar(50)			Department of the expenses
Created_at	Timestamp	NO		Create of expenses
Updated_at	Timestamp			Update of expenses

Table 3.2.8: Expenses Table of Coconut Processing Monitoring System.

Customer

Custumers	admin			
Attribute	**Type**	**Null**	**Default**	**Description**
Customer_id	INT(25)	NO	PK	Id of the customer
Name	VARCHAR(100)	NO		Name of the customer
Contact	Text	NO		Contact of the customer
Created_at	Timestamp	NO		Create of the customer
Updated_at	Timestamp	NO		Update of the customer

Table 3.2.9: customer Table of Coconut Processing Monitoring System.

System_config

System_config	admin			
Attribute	**Type**	**Null**	**Default**	**Description**
Config_id	Int(11)	NO	PK	Id of the config
Config_key	Varchar (100)	NO	Unique	Key of the config
Config_value	Text			Value of the config
Config_type	Enum	NO		Type of the config
Description	Varchar(255)	NO		Description of the config
Created_at	Timestamp			Create of system
Updated_at	Timestamp			Update of the system

Table 3.2.12: System_configTable of Coconut Processing Monitoring System.

Users

Users	admin			
Attribute	**Type**	**Null**	**Default**	**Description**
Users_id	Int(11)	NO	PK	Id of the user
Name	Varchar(100)	NO		Name of the user
Email	Varchar(100)	NO	Unique	Email of the user
Role	enum			Role of the user
Password_hash	Varchar(255)			Pass of the user
Created_at	Timestamp	NO		Created at the user
Updated_by	Timestamp			Update of the user

Table 3.2.11: Users Table of Coconut Processing Monitoring System.

Material_withrawal

Material_Withdrawal	**admin**			
Attribute	**Type**	**Null**	**Default**	**Description**
Withdrawal_id	Int (11)	NO	PK	Id of the material
Inventory	Int (11)	NO	Unique	Inventory of the material
Requested_quatity	Decimal(10,2)	NO		Request of withdrawal
Batch_id	Int (11)	NO	Unique	Batch of material
purpose	Int (11)	NO	Unique	Purpose of withdrawal
Status	Text	NO		Status of material
Requested	Enum	NO		Request of the material
Approved_by	Int (11)	NO	Unique	Approve_by withdrawal
Approved_at	Timestamp	NO		Approve_at withdrwal
Rejection_reason	Text	NO		Rejection of withdrawal

Table 3.2.10: material withdrawal Table of Coconut Processing Monitoring System.

Network Design

A network design outlines the architecture that supports communication between various devices. The network diagram below depicts a Wireless LAN

(WLAN) configuration where multiple devices connect to a central WiFi router. In this design, the router serves as the hub, providing wireless internet access to all connected devices. These devices, including computers, establish direct wireless connections to the central server, forming the core communication layer within the network and ensuring uninterrupted access for users.

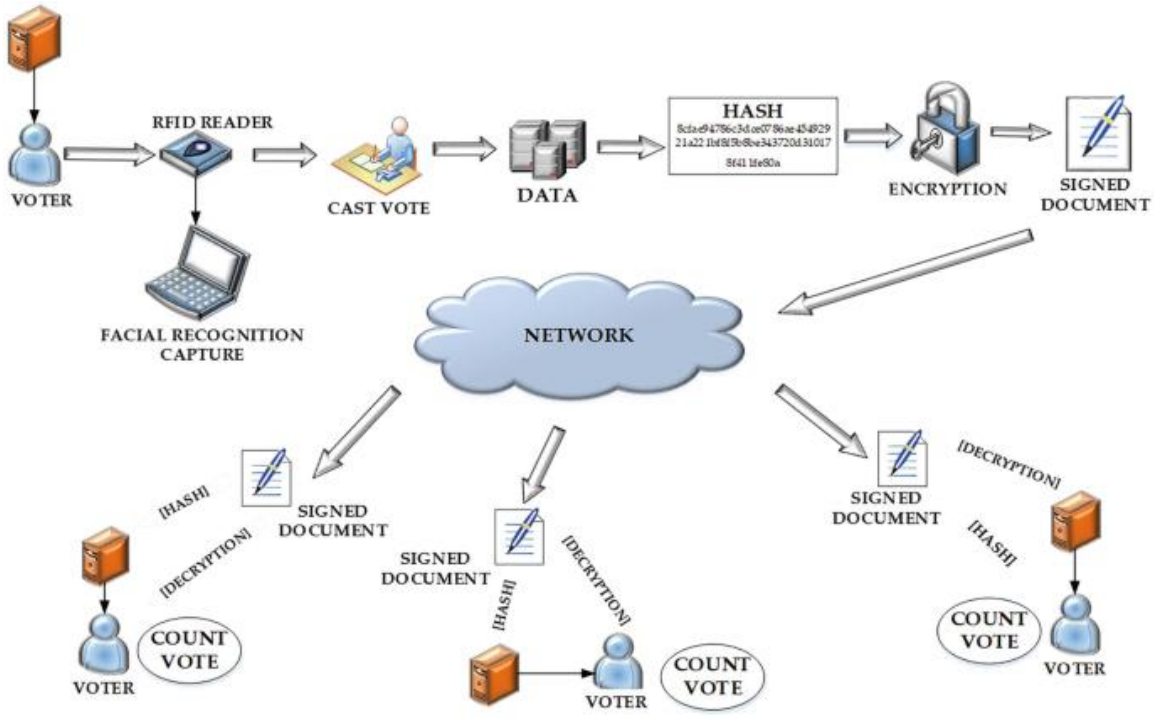


Figure 3.9.0: SS

This figure shows the network design

Network Topology

The network topology implemented for this system is a star topology, where all devices are connected to a central node, which is the Wi-Fi router. In a star topology, each device (node) communicates indirectly through the central hub, ensuring that devices connect directly to each other. The server and admin have primary access to control the network settings, managing devices connected via the Wi-Fi router. Although typically implemented with wired Wireless Area Network (WAN) connections, this system uses wireless connections (WLAN) for user devices, while the administrator and core services can still be wired for enhanced reliability.

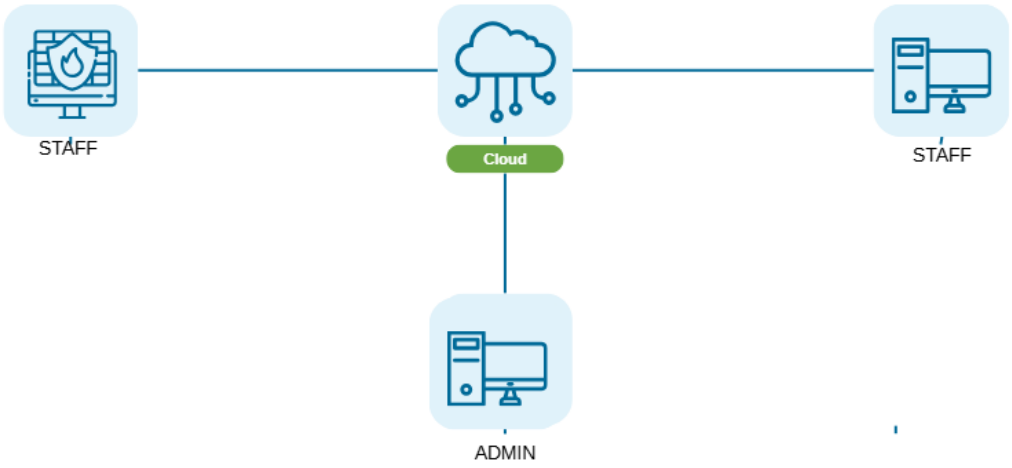


Figure 3.9.1: Network Topology of Coconut Processing Monitoring System. This figure shows the network topology

Technology Stack Development

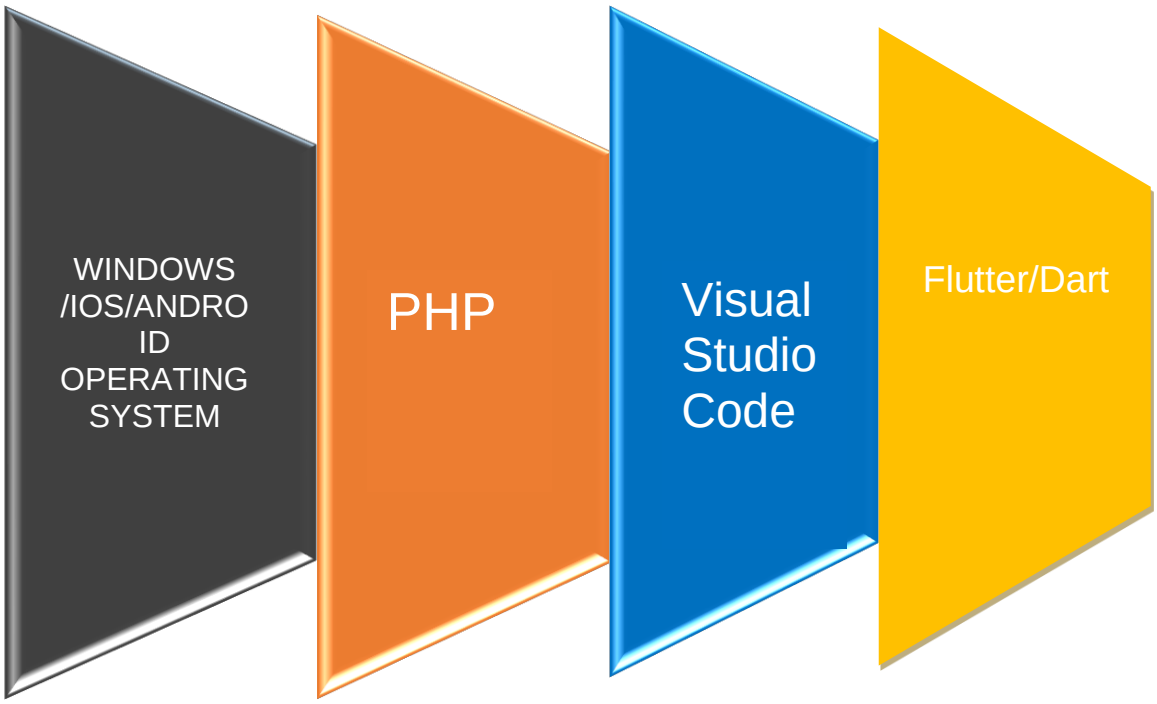


Figure 3.9.10: Technology Stack Development of Coconut Processing Monitoring System.

Software Specification

Software Requirements	
Language	Php, Flutter
Technology	Visual Studio Code
Database	MySQL
IDE	Visual Studio Code
Operating system	Microsoft Windows 11

Table 3.3.0: The Software Specification Table for Coconut Processing Monitoring System.

Hardware Specification

Processor	Intel Core i5-11400H CPU @2.70GHZ
Solid State Drive	250gb
Ram	8gb RAM

Table 3.3.1: The Hardware Specification Table for FACCALTI System: Faculty Call for Time System.

Program Specification

Language	Php
Database	MySQL, Wamp Server

Table 3.3.2: The Software Specification Table for FACCALTI System: Faculty Call for Time System.

List of Modules

Production Management Module – This module monitors and controls the entire coconut processing workflow, including scheduling, batch tracking, machine utilization, and production output recording.

Inventory Management Module – This module manages raw materials, work-in-progress items, and finished goods by tracking stock levels, movements, and reorder alerts.

Quality Assurance Management Module – This module ensures product quality and compliance by recording inspections, monitoring defects, and maintaining quality control documentation.

Worker Monitoring Module – This module tracks worker attendance, task assignments, shift schedules, and productivity performance throughout the production process.